# How College Enrollment Limits Household Responses to Work Incentives: Evidence from a Cash Transfer Experiment*

Li Gan, Xiongfei Guo, Qing He, Pengzhan Qian and Junhui Wang†

April 21, 2026

**Abstract**

This paper evaluates the first cash-transfer experiment with work incentives in a developing-country setting, in which households received monthly cash rewards proportional to their earnings for one year. We find that the program effectively increases households' labor supply, consumption, and savings, with substantial heterogeneity across household types. Households with college students primarily increased food and education expenditure, but showed only a limited increase in labor supply, potentially because the child was away at college. Households with younger children increased labor supply and focused on saving, likely in anticipation of future college expenses. Distinct from findings in developed countries, these patterns suggest that households prioritize resources for their offspring rather than investing in productivity. To formalize these mechanisms, we develop a two-period model that illustrates how these factors shape the observed patterns.

JEL: I38, J22, C93, O12

Keywords: Work incentive, field experiment, consumption, saving, education

*The study was approved by Human Research Protection Program (HRPP) at Texas A&M University under protocol number IRB2019-1069M. This RCT was registered as AEARCTR-0005212 (Gan et al., 2020).

Xiongfei Guo acknowledges support from the National Natural Science Foundation of China (Grant No.72403046). Qing He acknowledges support from the National Social Science Foundation of China (Grant No.20CJL018). Junhui Wang acknowledges support from the National Natural Science Foundation of China (Grant No.72073108). All funders had no role in the design of the study, the collection, analysis, or interpretation of data, the writing of the manuscript, or the decision to submit the article for publication.

We are grateful to Yonghong An, Songyu He, Manuel Hoffmann, Keyang Li, Yi Lu, Roberto Mosquera, Jiee Zhong, and Xuanli Zhu for their insightful comments and guidance.

†Gan: Texas A&M University (email: ganli@tamu.edu). He: Southwestern University of Finance and Economics (email: heqing@chfs.cn). Qian: Royal Holloway University of London (email: Pengzhan.Qian@rhul.ac.uk). Wang: Southwestern University of Finance and Economics (email: wangjunhui@chfs.cn).

Corresponding Author: Xiongfei Guo (email: guoxiongfei@uibe.edu.cn)

# 1 Introduction

Over the past decades, cash transfer programs have become among the most prominent anti-poverty initiatives in developing countries (Aber and Rawlings, 2011; Banerjee et al., 2017; Hanna and Olken, 2018). Yet many of these programs lack a work-incentive component[1] (Table 1). Such a feature may be particularly relevant in developing contexts, as it encourages households to invest immediately in productivity (Goodman-Bacon and McGranahan, 2008; Card et al., 2010; Baird et al., 2016; Card et al., 2018; Fisher and Rehkopf, 2022; McIntosh and Zeitlin, 2024, 2025). However, households in developing countries may instead prioritize current consumption or saving for future needs due to limited investment opportunities and uncertain returns.

To examine how households respond to work incentives in developing settings, the China Household Finance Survey (CHFS) at Southwestern University of Finance and Economics (SWUFE) conducted a field experiment between 2014 and 2017. The experiment took place in Wutongqiao, an urban district of Leshan City in Sichuan Province, China. A total of 251 households (130 households treated), comprising 727 individuals, were randomly assigned to treatment and control groups. Households in the treatment group received monthly cash incentives over a one-year period, structured through a tri-phasic reward scheme modeled after the Earned Income Tax Credit [2] in the U.S. for its explicit benefits to productivity enhancement. The EITC has played a central role in poverty alleviation for more than five decades and is the most extensively studied anti-poverty program in the U.S.[3], making it a valuable benchmark for comparison. Importantly, unlike the EITC, which operates through the tax system, our program delivered cash transfers directly to households, thereby avoiding

[1]We define work-incentive component broadly as any feature that provides households with financial incentives to work.

[2]The Earned Income Tax Credit (EITC) is a refundable tax credit in the United States designed to encourage work and support low-income earners. Its benefit follows a tri-phased structure, varying with income as it rises (see Figure 2 for example). The design is incentive-compatible, as it encourages employment by increasing the returns to work.

[3]For comprehensive reviews of the EITC, see Hotz and Scholz (2001), Meyer and Rosenbaum (2001), Eissa and Hoynes (2006), Hoynes and Rothstein (2016), and Nichols and Rothstein (2016). For a historical perspective, refer to Ventry (2000).

reliance on individual income tax systems that developing countries typically lack.

Consistent with studies from developed countries, our results show that program participation significantly increased household labor supply at the extensive margin. The number of workers per household rose by 0.19 persons (23%), leading to a 41-hour (33%) increase in total household hours worked and a 329 RMB (43%) increase in monthly household earnings. On the consumption side, participating households raised their monthly spending on food by 206 RMB (23%) and on education by 223 RMB (358%). They also became 11 percentage points more likely to report holding bank deposits.

However, we find that increases in income do not translate into higher consumption. These effects are driven by two distinct groups of households. Specifically, households with college students increased spending on food and education while maintaining their existing levels of labor supply and earnings. In contrast, other households substantially increased labor supply and earnings but did not raise their consumption, choosing instead to increase savings. Our analysis reveals that these patterns are driven by labor-capacity constraints and child-oriented investment priorities rather than by incentives to work. In a developing setting, a child is often regarded as a worker rather than a full-time student. When a household member attends college, the family not only loses a worker but the remaining earners must maintain full employment to offset the loss of income and meet education expenses. These households may wish to increase labor supply but are constrained by workforce capacities. In contrast, other households with greater capacity choose to work and earn more, saving the additional income for their children. To formalize this mechanism, we develop a two-period model in which households choose the number of workers and savings to maximize utility, with one household member (the child) leaving for college in the second period. The model maps household decisions to the work incentive and reproduces behavioral patterns consistent with our reduced-form estimates.

While many developing countries have implemented cash transfer programs, most of the existing programs impose conditionalities such as school attendance[4] or health checkups.[5] As expected, studies have consistently showed that these programs increased household consumption,[6] particularly on food[7]. However, because these conditionalities do alter the return of work or investment, the evidence on labor supply is mixed,[8] and only few have found evidence of enhancement in productivity.[9]

Our study contributes to the literature by providing the first experimental evidence on how cash transfer programs with EITC-style incentives operate in a developing-country context. We demonstrate that work incentives originally designed within tax systems can be effectively replicated through direct cash transfers. These transfers significantly increase education expenditure and savings, both of which are critical for fostering long-term poverty alleviation. Unlike the existing literature, we find that the effects on consumption and saving are primarily driven by investment in children's human capital. Moreover, our evidence shows that pursuing higher education—a major investment in human capital—imposes severe constraints that limit households' ability to adjust their labor supply. This suggests a strong connection between labor supply and human capital investment in developing countries.

Our findings have several important implications for the design of work-incentive programs in developing countries. First, we find program's effects on food and education expenditures that are comparable to those of existing CCT programs in developing countries, which also promoting labor participation. This underscores the potential of integrating

[4]See Skoufias and McClafferty (2001); Skoufias (2001); Maluccio and Flores (2005); Attanasio and Mesnard (2006); Chaudhury and Parajuli (2010); Fernandez and Olfindo (2011) for related studies.

[5]See Gertler (2000); Gertler et al. (2001); Skoufias (2001); Fernald et al. (2008, 2009); Fernandez and Olfindo (2011) for related studies

[6]See Maluccio (2005); Maluccio and Flores (2005); Masino and Niño-Zarazúa (2020) for related studies.

[7]See Skoufias and McClafferty (2001); Skoufias (2001); Hoddinott and Skoufias (2004); de Bem Lignani et al. (2011); Attanasio and Mesnard (2006); Martins and Monteiro (2016); Todd et al. (2020) for related studies.

[8]See Blattman et al. (2012, 2014); Barrientos and Villa (2015); Blattman et al. (2020); Gerard et al. (2021); Purba et al. (2023); Laguinge et al. (2024); Yuliani and Nasrudin (2024) for studies that find positive impacts. See Ribas and Soares (2011); Baird et al. (2018); Fruttero et al. (2020); Dona (2023); Vidigal (2023) for studies that find mixed results. See Samaniego and Tejerina (2010); Amarante et al. (2011); Garganta and Gasparini (2015) for studies that find negative impacts.

[9]Skoufias (2001) found CCT program has substantial effect on lifetime productivity. Angelucci et al. (2012); Masino and Niño-Zarazúa (2020) found increase in savings. Ahaibwe et al. (2013); Alix-Garcia et al. (2013); Todd et al. (2020) found increase in farming inputs such as seeds and land.

incentive-based designs into existing programs to enhance targeting efficiency and reduce implementation costs. In addition, the unique developing-country context of our study reveals that low-income households prioritize saving for child investment. Households with college students, despite their strong potential for upward mobility, benefit less from the program as their labor supply is less elastic. These results highlight the need for policies that alleviate such constraints and call for policymakers to recognize complementarities across social programs, designing safety nets as integrated systems rather than standalone interventions.

The remaining of the paper is structured as follows: Section 2 provides background information, including the timeline, pilot study, and experiment design. Section 3 presents baseline results for labor supply, earnings, consumption, saving as well as health metrics. Section 4 compares the heterogeneity between households with and without college students. Section 5 explores mechanisms. Section 6 concludes.

# 2 Timeline and Experimental Design

To examine whether EITC-style incentives can be implemented through cash transfers, the Center for China Household Finance Survey (CHFS) at Southwestern University of Finance and Economics conducted the *Labor Income Reward Program* (LIRP) experiment between 2014 and 2017. The study took place in Wutongqiao, an urban district of Leshan City in Sichuan Province, China. To contextualize the experimental site, Wutongqiao is a typical mid-tier urban district in a developing region of China. The district has a population of about 310,000, roughly equivalent to that of Pittsburgh, and a per capita GDP of 6,713 RMB (82% of the national average) or 1,068 USD (all dollars mentioned are in 2015 USD).[10,11] In 2014, we estimated that approximately 23.6% of households in this area were recipients of

[10] Wutongqiao District Government. (2016). Statistical Report of the Wutongqiao on the Development of Social Services in 2015. http://www.wtq.gov.cn/wtqq/tjxxwtq/202005/7c255f4d8af645a4b9bd4b99d54f6252.shtml

[11] World Bank. (2023). GDP per capita (constant 2015 USD) - China. https://data.worldbank.org/indicator/NY.GDP.MKTP.KD?locations=CN

government aid.[12] The district provides a relevant setting for examining program impacts in a developing-country context.

Figure 1 illustrates the primary phases of the study: the initial phase consisted of a pre-pilot household survey conducted in May 2014, followed by a pilot experiment from December 2014 to May 2015, and, finally, the main experimental phase, which took place from October 2015 to January 2017.

## 2.1 Pre-pilot Survey

The pre-pilot survey employed a three-stage stratified random sampling technique to select 808 households from the experimental site and collected a wide range of socioeconomic indicators, with particular attention to work status and earnings. Its primary purpose was to calibrate the intervention parameters (e.g. phase rates and maximum reward) to ensure budget feasibility.

## 2.2 The Pilot

For the pilot study, participants were recruited from the pre-pilot sample, excluding households with per capita monthly earnings above 1,200 RMB (the sample average) and those without working-age members. We then used a logistic regression model to estimate the probability of employment for each unemployed individual, controlling for age, marital status, gender, pension eligibility, and household registration. Based on predicted employability and willingness to participate, the final sample included 59 households, of which 27 were randomly assigned to the treatment group.[13]

Table 2 displays the covariate balance between the treatment and control groups. For individual characteristics, we focus on able-bodied individuals and include factors such as

[12] At the time, China offered only means-tested programs. In the experiment area, the most common were the Minimum Living Standard Guarantee (Dibao) and Disability Assistance.

[13] Although this selection strategy limits external validity, internal validity is preserved through random assignment. Importantly, this selection procedure was applied only in the pilot phase to identify implementation issues prior to the main experiment.

gender, age, education, labor supply, and earnings. For household characteristics, we consider variables describing household composition and poverty status. For each variable, we calculate the group mean and standard deviation for the treatment group (first column) and control group (second column). The third and fourth columns show the mean differences between the groups and p-values from a two-sample t-test. The lack of statistically significant differences across individual and household characteristics indicates successful randomization.

All households in the treatment group were enrolled in the program, while households in the control group received no financial incentives. Importantly, eligibility for other welfare programs was unaffected for both groups. A baseline survey was conducted in December 2014, followed by an information session for all treatment households to explain the scheme's details and timeline. The intervention then began, and a follow-up survey was carried out in March 2015.

Figure 2 illustrates the reward scheme used in the pilot study, which includes three distinct phases: phase-in, plateau, and phase-out. The scheme parameters were calibrated based on local income levels to create meaningful work incentives while maintaining budget feasibility. Both the phase-in and phase-out regions apply an identical phasing rate of 50%, modeled after EITC in the U.S. The maximum reward per capita is set at 250 RMB per month (40 USD), with a break-even point of 1,200 RMB (194 USD). For the experiment, we adopted a similar intervention, with a slight modification to account for the number of children in the household, which we discuss later.

To claim the reward, participating households were required to substantiate their reported earnings with documented evidence. To validate the materials, a dedicated team, comprising CHFS staff and student assistants, conducted employer verification calls for employed participants. For self-employed individuals, a field team performed random site visits at bi-weekly to tri-weekly intervals, further verifying income information through neighborhood inquiries and cross-checking against similar occupations.

After verification, the reward amount was determined through the following procedure: households' total earnings from the previous month were normalized by family size to calculate per capita earnings.[14] This per capita figure was then applied to the scheme in Figure 2 to determine the per capita reward. The total household reward was calculated by multiplying the per capita reward by the family size. Rewards were then transferred to the household head's bank account in the middle of that month.

For example, consider a household with two parents and one child, giving a family size of three. If the household's total earnings is 1,500 RMB in the previous month, the per capita earning would be $1,500 \div 3 = 500$ RMB. This per capita earning corresponds to a per capita reward of 250 RMB, resulting in a total reward of $250 \times 3 = 750$ RMB.

## 2.3 The Main Experiment

For the main experimental phase, which began in October 2015, we recruited additional households through community referrals.[15] Within this pool of households, we calculated per capita earnings for each household. Households with monthly per capita earnings exceeding 1,000 RMB—the break-even point of the new scheme—or those unwilling or unable to work were excluded from the sample. The remaining households were stratified by per capita earnings and then randomly assigned to treatment and control groups. The local authority considered it unfair to keep some of the poorest and most hard-working households in the control group; at their request, approximately 20% of control households were reassigned to the treatment group. The final sample comprised 533 individuals (284 treated) from 192 households (103 treated).[16]

[14] Only spouses, parents, dependents, and household members who have lived together and shared expenses for more than six months were considered household members.

[15] This approach was necessary because the pool of eligible participants identified in the pre-pilot survey had been exhausted, and public recruitment is generally not permitted in China. Although community referrals may limit external validity, internal validity remains intact, as households were randomly assigned to treatment and control groups. We will fully discuss the implications for external validity at the end of the paper.

[16] While modest in size by survey standards, this sample is typical for resource-intensive field experiments with frequent monitoring in developing-country settings. The randomized design and bi-monthly panel

As in the pilot phase, a baseline survey was conducted in October 2015, followed by an information session for the treatment households before the intervention began. Five follow-up surveys were then administered on a bi-monthly basis throughout 2016.

Table 3 presents the covariate balance between the treatment and control groups at the initial assignment (left panel) and actual assignment (right panel). For the initial assignment, mean differences across all variables are statistically insignificant, indicating successful randomization. However, in the actual assignment, two-sample t-tests show that treatment group households have a monthly per capita income 133.9 RMB lower than the control group, significant at the 5% level. Moreover, households in the treatment group work 35.7 fewer total hours per month, and their monthly total earnings are 235.8 RMB lower, with both differences statistically significant at the 10% level.

Given the compromised randomization, Ordinary Least Squares (OLS) estimates are likely biased. To address this issue, we employ an Instrumental Variables (IV) approach, using the initial assignment as an instrument for actual treatment status. This approach is valid because the initial assignment is strongly correlated with actual treatment (relevance condition) but affects outcomes only through its effect on treatment status (exogeneity condition). We also take the advantage of panel data and control for both household and time fixed effects in case of other endogeneity. The first-stage and reduced-form regressions for baseline analysis are specified as follows:

$$\text{First-stage:} \quad T_{i,t} = \alpha_i + \lambda_t + \beta D_{i,t} + \epsilon_{i,t} \tag{1}$$

$$\text{Reduced form:} \quad Y_{i,t} = \alpha_i + \lambda_t + \delta D_{i,t} + \xi_{i,t} \tag{2}$$

where $T_{i,t}$ represents the actual treatment status for household $i$ at time $t$. $D_{i,t}$ denotes the initial treatment assignment. $Y_{i,t}$ consists outcome variables that measure labor supply,

structure provide internally valid estimates of program impacts, ensuring credible identification despite the sample size.

earnings, consumption and saving. $\alpha_i$ and $\lambda_t$ represent household and time fixed effects, respectively, while $\epsilon_{i,t}$ and $\xi_{i,t}$ denote the error terms. Note that all time-invariant household characteristics are absorbed by the household and time fixed effects. To ensure the robustness of our results, we compare the IV estimates with simple OLS regressions based on both initial and realized treatment status. We also re-estimate the baseline specifications after excluding switcher households. These exercises demonstrate the robustness of the baseline results and ensure that treatment switches do not drive the findings. Finally, we report estimates using household-level data to capture aggregate effects.

The scheme used in the main experiment closely follows that of the pilot study but introduces a tiered structure based on the number of children in the household (Figure 3). The break-even point is set at 1,000 RMB, lower than the previous threshold of 1,200 RMB. For households with fewer than two children, the maximum reward is 200 RMB per capita per month, with phase-in and phase-out rates remaining at 50%. In contrast, for households with two or more children, the maximum reward per month is reduced to 160 RMB per capita (25 USD[17]), and the phase-in and phase-out rates are set at 40%.

This tiered structure accounts for the additional support needed by larger families while ensuring a balanced distribution of rewards. The decision to vary benefits by family size was inspired by the EITC's similar approach, though our implementation uses per capita calculations rather than total household figures. The calculation process for the total reward remains unchanged.

## 2.4 Compliance and Income Validation

The average program participation rate remained high at 95.3% throughout the intervention period, and the average claiming rate was 84.6%. During the study, 18 households withdrew for various reasons, including exceeding the income threshold or violating legal guidelines. As we demonstrate later, the baseline results are robust to this level of program attrition. To

[17] In 2015 U.S. Dollars

assess the validity of reported income, Figure 4 compares the estimated density distributions of reported earnings (solid line) and verified earnings (dashed line). Kolmogorov–Smirnov and Anderson–Darling tests show no significant differences (p-values of 0.83 and 0.91, respectively), confirming the accuracy of the income measure and indicating minimal scope for manipulation.

# 3 Baseline Results

In this section, we will first briefly show that the program increased household labor supply and earnings as intended. Overall, we find consistent evidence that the program significantly increased number of workers in the household, household monthly hours worked, and household monthly earnings. Next, we examine program's effects on a variety of outcomes such as consumption and saving as well as health metrics. Overall, we find program participation has significantly increased household food consumption and education expenditure. In addition, treated households are also more likely to report having deposit in their bank accounts.

## 3.1 Impacts on Household Labor Supply and Earnings

Classic theory suggests that EITC-style incentives primarily increase labor supply at the extensive margin. To reflect this, we measure household labor supply by the number of members who worked in the previous month. We also calculate the total hours worked by households, which captures changes at both the extensive and intensive margins. Our measure of household earnings includes income from hired and self-employed jobs, income from household-owned small businesses, and farming income in the previous month. The program reward is not counted as earnings.

Table 4 reports the IV estimates for household labor supply and monthly earnings. Panel A shows the estimates from the pilot study. Results show that treatment increases the number of workers by 0.35 person. It also increases monthly total hours worked by 95.64

hour per household. As for earnings, the pilot treatment increases household's monthly total earning by 349.73 RMB. However, the effect on earning is statistically insignificant. Panel B shows the estimates from the main experiment. Results show that treatment increases the number of workers by 0.19 person. It also increases monthly total hours worked by 41.03 hour per household. As for earnings, the treatment increases household's monthly total earning by 329.53 RMB. All results are statistically significant at 5% level. The high first-stage F-statistics indicates strong instrument relevance in the first stage, supporting the validity of our IV approach. To summarize, we find that program participation significantly increased household labor supply at the extensive margin, thereby raising household earnings.

## 3.2 Impacts on Consumption and Saving

As previously discussed, work-incentive programs are expected to affect both household expenditures and savings. For consumption, we examine monthly household expenditures across various categories, including food, eating out, utilities, sundry items, transportation, communication, clothing, appliances, education, and healthcare.

Figure 5 reports the IV estimates for each consumption category. As shown in the figure, treated households increased food consumption by 206 RMB per month and education expenditure by 233 RMB per month. Both effects are statistically significant at the 5% level, while effects on other categories are small and statistically insignificant.

For savings, we examine the program's effect on the likelihood that households report having bank deposits[18]. Figure 6 presents the results. As shown in the figure, treated households from the experiment were 11 percentage points more likely to report having deposits in their bank accounts with statistical significance at the 10% level.

To summarize, we find that program participation significantly increased household food consumption and education expenditure. Participants were also significantly more likely to report having deposits in their bank accounts.

[18] We measure saving in this way due to the low response rate for the exact balance in household bank accounts.

### 3.3 Impacts on Health and Productivity

Increased food consumption may improve health outcomes through better nutrition or enhance productivity by reducing sick leave. To test these mechanisms, we use the initial treatment assignment as an instrument for food consumption in the previous period and examine the program's effects on individuals' self-reported symptoms, including fever, body pain, diarrhea, cough, and palpitations, as well as on the severity of symptoms and the probability of working.

Figure 7 presents the estimated program effects on health and labor productivity through food consumption. Overall, we find no evidence that improved food consumption leads to better health or labor productivity.

Work-incentive programs may also create financial incentives for households to invest in productivity, such as purchasing better seeds to increase crop yields. Accordingly, the program may have encouraged households to finance such investments via intra-household transfers. To test this mechanism, we examine whether the program affected the probability of giving or receiving cash gifts. However, we find no evidence of such changes (Figure 6).

Table 5 summarizes the key findings from the baseline analysis (left panel). We find that program participation significantly increased labor supply, earnings, food consumption, education expenditure, and savings. However, we find no evidence that the increase in food consumption improved health outcomes or labor productivity, nor do we find evidence of increased inter-household transfers to finance productivity-related investments.

### 3.4 Robustness Checks

In this section, we address concerns about potential bias from imbalanced treatment assignment by comparing our baseline IV estimates with four additional specifications (Figure 8). First, we report intention-to-treat (ITT) estimates, obtained by regressing each main outcome on the initial assignment indicator. These effects are causal in the presence of noncompliance and represent the reduced-form estimates corresponding to our IV results. The

ITT estimates (blue) closely track the baseline IV estimates (brown) but are, as expected, somewhat smaller because noncompliance attenuates the treatment effect.

Next, we present OLS estimates using actual participation. These results (red) are similar to the baseline IV estimates (brown) but somewhat larger, suggesting positive selection into participation—that is, switcher households exhibit stronger responses.

To address this issue, we estimate OLS regressions restricting the sample to non-switcher households. The results (black) remain close to the baseline IV estimates (brown), indicating that the main findings are not driven by switchers.

Finally, we assess the potential impact of sample attrition. Eighteen households exited the program prematurely for various reasons, yet they continued to be tracked in the survey. Including these households may introduce bias. To evaluate this possibility, we re-estimated the baseline regressions after excluding the attrited households. The results (green) remain consistent to the baseline estimates (brown), confirming that program attrition does not affect our findings.

Together, these exercises demonstrate that our results are not driven by estimation strategies or sample composition.

# 4 College-student Households

In this section, we first show that the treatment effect on education expenditure was primarily driven by households with members attending college during the experiment (*college-student households*, henceforth), rather than households without college students (*non-college-student households*, henceforth). We then compare key outcomes such as labor supply, earnings, consumption, and saving between these two types of households to highlight their distinct behavioral patterns. Overall, our analysis reveals that *college-student households* primarily drive the effects on food and education expenditures, whereas *non-college-student households* account for the effects on labor supply, earnings, and saving.

While our survey did not ask about the specific purposes of spending or children's schooling, we can infer the nature of education expenditures by examining children's education stages based on their ages. Table 6 presents the age distribution and corresponding education stages (in parentheses) for children in the treatment and control groups. Specifically, we construct indicator variables for the presence of children within each age group. The first two columns of the table report the means and standard deviations (in parentheses) of these variables, the third column reports the mean differences and pooled standard deviations from a two-sample t-test, and the last column reports the associated p-values. The results show no meaningful differences between the treatment and control groups across education stages, suggesting that the observed effect on education expenditure is not driven by differences in child composition between the two groups.

We next focus on households with children at different stages of education and examine the heterogeneity in treatment effects on education expenditure. Specifically, for each age group defined in Table 6, we restrict the sample to households within that group and compare the results with those from the remaining sample. Figure 9 shows that *college-student households* are the primary drivers of the treatment effect on education expenditure. Program participation increased education expenditure by 320 RMB in March and by 2,676 RMB in September, both significant at the 5% level. For the remaining months, the estimated effects are small and statistically insignificant. Although our survey does not record the specific purposes of these expenses, the timing of the effects suggests that they are tied to the academic semester. Moreover, our mechanism analysis indicates that these expenses are most likely non-deferrable, which we discuss in later sections.

Motivated by this pattern, we examine whether *college-student households* differ systematically from other households in their broader responses to the program. Specifically, we compare the dynamic treatment effects to understand how responses evolved differently over time. Overall, we find that the two groups exhibited distinctly different effects.

To begin with, the labor supply effects are entirely driven by *non-college-student house-*

*holds* at the extensive margin. Figure 10 plots the dynamic treatment effects on the number of workers relative to the pre-treatment period. Among *non-college-student households* (red), the number of workers rose substantially within two months of program implementation and remained elevated throughout the experiment. The estimated effects range from 16 to 37 percentage points and are statistically significant at the 5% level in most periods. By contrast, *college-student households* (blue) exhibit no significant change at any point.

Similarly, the effects on monthly hours worked are also driven by *non-college-student households*. Figure 11 plots the dynamic treatment effects on total hours worked. Among *non-college-student households* (red), the estimated effects range from 30 to 80 additional hours per month per household and are statistically significant at the 5% level in most periods. By contrast, *college-student households* (blue) show no significant change in any period.

Consequently, only *non-college-student households* experienced an increase in monthly earnings. Figure 12 plots the dynamic treatment effects on monthly household earnings. Among *non-college-student households* (red), the estimated effects range from 231 RMB to 405 RMB per month per household and are statistically significant at the 5% level in most periods. By contrast, *college-student households* (blue) show no significant change in any period.

As for consumptions, we find that the effects on food and education expenditures are both concentrated among *college-student households*. Figure 13 plots the dynamic treatment effects on household food consumption. Among *college-student households* (blue), the estimated effects on food consumption range from 269 RMB to 589 RMB per month per household and are statistically significant at the 5% level in most periods. By contrast, *non-college-student households* (red) show no significant changes in any period.

With earnings increasing but consumption remaining unchanged, the effects on savings should be primarily driven by *non-college-student households*. Figure 14 plots the dynamic treatment effects on the probability that a household reports having bank deposits. Among *non-college-student households* (red), the estimated effects range from 1 to 20 percentage points and are statistically significant in most periods. By contrast, *college-student households* (blue) show no significant changes in most periods, except for November.

To summarize, among *college-student households*, the program led to significant increases in food and education consumption but no meaningful changes in labor supply, earnings, or savings. By contrast, *non-college-student households* significantly increased labor supply, earnings, and savings but showed no increase in consumption. Table 5 presents a comparison of these results (right panel).

# 5 Mechanisms

To understand the factors driving these patterns, we first compare *college-student households* with *non-college-student households* using baseline data. Specifically, we estimate a Probit model in which a binary indicator for being a *college-student household* is regressed on a set of household characteristics, including family size, the numbers of senior members (age>65) and children (age<16), the amount of government aid, income per capita, the numbers of workers and able-bodied members, monthly hours worked; the highest education level, whether the household has a bank deposit, and whether the household received a cash gift in the previous month.

Figure 15 highlights the difference between *college-student households* and *non-college-student households*. These factors can be summarized into three dimensions. First of all, *college-student households* tend to have fewer workers and able-bodied members, and their existing workers tend to work substantially longer hours. Secondly, households with more senior members and fewer children tend to be *college-student households*. Thirdly, *college-student households* are more likely to have bank deposit in the baseline period. We will discuss how these aspects shaped the heterogeneity between *college-student households* and *non-college-student households* in the following sections.

## 5.1 Labor-capacity Matters More than Incentives

We begin by explaining why *college-student households* did not respond to the work incentives. First, the Probit estimates show that *college-student households* and *non-college-student households* have similar income per capita and the amount of government aid (Figure 15). This suggests that *college-student households* are not particularly poorer than others and, therefore, are not more likely to fall within the phase-in region where marginal rewards are strictly positive. In other words, they are not differently incentivized. To test this, we re-estimate the treatment effect on labor supply using baseline regression with interaction term between a binary variable indicating households' baseline positions in the schedule and their treatment status. The result shows that households residing in the phase-in region did not respond differently in terms of number of workers compared to others (first column of Table 7). Therefore, differences in incentives alone do not explain the heterogeneity in labor-supply effects.

Secondly, Figure 15 shows that *college-student households* have significantly fewer able-bodied members. Because teenagers leave the household upon entering college, these households effectively have one fewer available worker than otherwise comparable households with the same family structure. Consequently, there are fewer working members in these households. Moreover, the remaining workers often have to work extra hours to offset the income loss associated with the lower labor capacity. As shown in Figure 15, *college-student households* are also significantly more likely to include members aged above 65. This pattern reflects China's tradition of intergenerational co-residence, where grandparents live with and support the family, allowing parents to focus on work and students to concentrate on studying. Importantly, this means that *college-student households* are more likely to operate near their maximum labor capacity, leaving limited scope to adjust their labor supply. To test this, we re-estimate the treatment effect on labor supply using baseline regression with an interaction term between a binary variable indicating whether the household had excess able-bodied members who were not working at baseline and the treatment status. The results show that the effect on number of workers is particularly pronounced among households with with available members to join the labor force. The coefficient on the interaction term is large and statistically significant at the 1% level (second column of Table 7). These findings suggest that differences in labor capacity between *college-student* and *non-college-student households* matter more than the incentives they face.

## 5.2 Education Expenditure in College-student Households

We next explain why *college-student households* did not increase savings but instead raised their consumption. As shown in Figure 15, *college-student households* are more likely to have bank deposits. This reflects the financial pressure imposed by college education. For a bachelor's degree at a public university in China, typical annual tuition fees for domestic students range from 4,000 to 6,500 RMB (approximately 566 to 920 USD), depending on the institution and major (Qian, 2025). This amount is roughly equivalent to 20%–35% of the annual income of households in our sample. Although tuition waivers and scholarships exist for low-income families, they rarely provide for travel and living expenses. To meet these costs, most college-going households were already accumulating savings before program participation. Consequently, the program had little impact on their saving behavior. To test this, we re-estimate the treatment effect on saving using baseline regression with an interaction term between a binary variable indicating whether the household reported saving at baseline and the treatment status. The results show that the effect on saving is substantially attenuated among households without prior deposits. The coefficient on the interaction term is large, negative, and statistically significant at the 1% level (third column of Table 7). These findings suggest that the program had no additional effect on savings among *college-student households*, as most were already saving to cover educational expenses.

As previously discussed, *college-student households* significantly increased food consumption after program participation. This pattern suggests these households used to face liquidity constraints, as they adjusted spending primarily toward essential goods once liquidity was eased. While the coexistence of savings and increased food consumption may appear contradictory, the two are in fact interrelated. Because tuition and related expenses are large relative to annual income, households prioritize saving over consumption, which tightens short-term liquidity and limits current spending. That is to say, households used to prioritize education expenditure and therefore limit their food consumption. Once they have additional income, they are able to relax on food consumption as well as providing more education expenditure.

## 5.3 Precautionary Saving in Non-college-student Household

Finally, we examine why *non–college-student households* focus primarily on saving. Since the earlier analysis suggests that *college-student households* prioritize college expenses, households with younger children may similarly prefer to save additional income for their children for similar purposes. To test this hypothesis, we compare the dynamic treatment effects between households with young children and those without any children.

Figure 16 plots the dynamic treatment effects on the probability that a household reports having bank deposits. Among households with children (blue), the estimated effects range from 26 to 39 percentage points and are statistically significant in most periods. By contrast, households without children (red) show no significant changes in any period. These results indicate that the increase in saving among *non–college-student households* is primarily driven by households with children, supporting our hypothesis that the additional saving reflects investment in children.

To summarize, our mechanism analysis indicates that the program had little effect on labor supply and saving among *college-student households*, primarily due to workforce shortages and pre-existing high savings associated with high college expenses. Similarly, the pronounced increase in saving among *non-college-student households*, consistent with our interpretation that households save in anticipation of future child-related expenses.

## 5.4 A Two-period Child Investment Model

As discussed, the heterogeneous responses of *college-student households* and *non-college-student households* are driven largely by their differing constraints rather than by the program's marginal incentives. We therefore propose a simple two-period framework to formalize this idea.

*Generic household problem.* The household faces a two-period discrete labor participation problem. $w_1$ and $w_2$ are the wages in the first and second periods, respectively. In the first period, the household has upon to $\overline{n}_1$ potential workers. It derives utility from a MaCurdy-type function combining consumption $C_1$ and the disutility of working, which depends on the number of participating workers $n_1$. $\rho$ governs the return of consumption, while $\gamma$ and $\theta$ determine the disutility of working. The household also saves $S$ for the second period, given the interest rate $r$.

In the second period, one worker (the child) is away for college. Therefore, the number of available workers falls to $\overline{n}_2 = \overline{n}_1 - 1$. The household values the child's utility in college and therefore provides education expenditure $E$. $\alpha$ and $\delta$ govern the utility from investing in the child.

$$u_1 = \frac{C_1^{1-\rho}}{1-\rho} - (\gamma n_1)^\theta \tag{3}$$

$$C_1 = w_1 n_1 - S \tag{4}$$

$$u_2 = \frac{C_2^{1-\rho}}{1-\rho} - (\gamma n_2)^\theta + \alpha E^\delta \tag{5}$$

$$C_2 = w_2 n_2 + rS - E \tag{6}$$

$$\overline{n}_2 = \overline{n}_1 - 1 \tag{7}$$

The household choose $n_1$, $n_2$, $S$, and $E$ to maximize its lifetime utility $U$, given discount factor $\beta$.

$$U = \max_{n_1, n_2, S, E} u_1 + \beta u_2 \tag{8}$$

*Non-college-student household.* In this framework, the *non-college-student household* is represented by the household in period one, while the college-student household is represented by the household in period two. When wages increase, we expect they to respond differently because they face different budget constraints and have different numbers of available workers.

For the *non-college-student household*, they will increase the number of workers to take advantage of the increased wage. However, as the household anticipates having fewer available workers and needs to pay child's education in the next period, it will refrain from increasing current consumption substantially and instead save more.

*College-student household.* For the *college-student household*, a wage increase also incentivizes it to provide more labor. However, because this household has fewer available workers, its labor supply response will be more limited than that of the *non-college-student household*. As the household cares about the child's education, it will transfer most extra income to the child[19].

To illustrate this mechanism, we conduct two experiments with a household of three members ($\overline{n}_1$=3). We first increase $w_1$ from 1 to 1.2 to mirror the field experiment. $w_2$ is unchanged as the field experiment is conducted in one year. We introduce a small wage variance with a normal distribution ($\sigma = 0.1$) and simulate the model 1000 times. Figure 17b shows that, given the parameters, the *non-college-student household* increases its total number of workers in response to the higher wage (blue solid line). It consumes slightly more (red dashed line) but saves most resources for the next period (green short dashed line).

For the second experiment, we increase $w_2$ from 1 to 1.2 while keeping $w_1$ unchanged

[19] In this two-period model, college-student households do not save. Although the model could be extended to a three-period setting in which education spending generates returns, such an extension is beyond our scope, as our reduced-form estimates capture only short-term effects.

(Figure 17a). As the number of workers starts at a relatively high level, the household cannot provide more labor (blue solid line). Given this inelastic labor supply, the total income can only rise proportionally with the wage. As predicted by the model, the household allocates most of the additional income to education (orange short dashed line). As shown in the figure, its consumption (red dashed line) has also increased by roughly 10%, confirming the empirical evidence.

In summary, the model replicates the key empirical patterns observed in the data and clarifies the mechanisms underlying them. Together with the empirical findings, these results suggest that the observed heterogeneous behavioral responses are largely driven by differing household constraints.

# 6 Conclusion and Discussion

Work-incentive programs are highly relevant in developing contexts because they creates financial incentives for households to invest in productivity. Yet, our understanding of the effectiveness of such approaches remains incomplete. By conducting the first experiment of its kind, we show that a cash transfer program with an EITC-style incentive can effectively increase food consumption, education expenditure, and savings, all of which are crucial for long-term poverty alleviation.

Our analysis indicates that constraints play a crucial role in determining the effectiveness of work-incentive programs in developing countries. For labor-supply responses, capacity constraints are central, as households cannot respond to incentives when they lack available workers or when existing workers are already operating at full capacity. For consumption and saving responses, it is largely driven by households prioritizing on saving for college education or children. With cash transfers, households only adjust essential consumption of food; otherwise, they prefer to save additional income. These contrasting mechanisms illustrate that constraints, rather than differences in how households perceive incentives, are key to explaining why incentive-based programs yield heterogeneous outcomes across different types of households.

Our findings have several important implications for policymakers in developing countries. First, the increases in food and education expenditures indicate that cash transfers with work incentives can generate effects similar to programs that directly target nutrition and education. This suggests that integrating incentive-based designs into existing transfer programs may improve targeting efficiency while simultaneously promoting work and avoid welfare dependency. Second, the heterogeneous effects on labor supply, consumption, and saving reveal that low-income households often face shortages of both labor and liquidity—two typical features of developing-country contexts. Importantly, these constraints are closely connected, and together they significantly limit the effectiveness of work-incentive programs. As a result, labor-constrained households benefit only modestly from the program and remain liquidity-constrained. A further lesson is that low-income households often place greater value on children than on current consumption or productivity investment. Taken together, these insights highlight the importance of coordinating work incentives with complementary policies that either relax these constraints or offer desirable investment opportunities. This calls for policymakers to view safety nets as integrated systems rather than standalone interventions.

For external validity, we follow the SANS framework—Selection, Attrition, Naturalness, and Scaling—proposed by List (2020). First, regarding selection, although participants were recruited through community referrals, the sample is representative of the intended target population—the working poor. All sampled households include able-bodied members and are substantially poorer than the local average, with average monthly household earnings of 876 RMB compared to 1,504 RMB locally. Second, attrition was minimal, at only 4.7%. For naturalness, the choice environment closely mirrored real-world decisions: households made genuine labor-supply choices in response to actual cash rewards. The strongest evidence of this comes from the official adoption of our design by the local government following the experiment, underscoring the program's realism and policy relevance. Finally, regarding scaling, while the findings are encouraging, caution is warranted. The small-scale design cannot capture potential implementation costs or equilibrium effects—such as labor-supply shocks or spillovers—that might arise if the program were expanded to the national level.

For future research, our study points to several promising avenues for further investigation. First, our experiment was conducted in a specific region of China, and its generalizability to other developing-country contexts needs further examination. Second, the one-year intervention period limits our ability to assess longer-term effects, including potential changes in household specialization patterns, human capital accumulation, or strategic adaptations as participants gain experience with the program. Third, studies that explicitly vary program parameters across treatment arms could further assess the effectiveness of designs with smaller or less frequent rewards, which is an important consideration for ensuring scalability in developing countries with tight budget constraints. Finally, our results highlight the link between human capital investment and household labor supply. This feature is not typically incorporated into standard household labor-supply models, yet it may be relevant for understanding the low college enrollment rates observed among low-income households around the world.

Ultimately, understanding how low-income households respond to work incentives is essential for designing transfer programs that both alleviate poverty and promote productive engagement. Future efforts to evaluate and refine such programs will benefit from incorporating these mechanisms into both policy design and empirical analysis.

# References


Aber, L. and Rawlings, L. B. (2011). *North-South knowledge sharing on incentive-based conditional cash transfer programs*. Citeseer.

Ahaibwe, G., Mbowa, S., and Lwanga, M. M. (2013). Youth engagement in agriculture in uganda: Challenges and prospects.

Alix-Garcia, J., McIntosh, C., Sims, K. R., and Welch, J. R. (2013). The ecological footprint of poverty alleviation: evidence from mexico's oportunidades program. *Review of Economics and Statistics*, 95(2):417–435.

Amarante, V., Manacorda, M., Vigorito, A., and Zerpa, M. (2011). Social assistance and labor market outcomes: Evidence from the uruguayan panes.

Angelucci, M., Attanasio, O., and Di Maro, V. (2012). The impact of oportunidades on consumption, savings and transfers. *Fiscal Studies*, 33(3):305–334.

Attanasio, O. and Mesnard, A. (2006). The impact of a conditional cash transfer programme on consumption in colombia. *Fiscal studies*, 27(4):421–442.

Baird, S., McIntosh, C., and Ozler, B. (2016). When the money runs out: do cash transfers have sustained effects on human capital accumulation? *World Bank Policy Research Working Paper*, (7901).

Baird, S., McKenzie, D., and Özler, B. (2018). The effects of cash transfers on adult labor market outcomes. *IZA Journal of Development and Migration*, 8(1):22.

Banerjee, A. V., Hanna, R., Kreindler, G. E., and Olken, B. A. (2017). Debunking the stereotype of the lazy welfare recipient: Evidence from cash transfer programs. *The World Bank Research Observer*, 32(2):155–184.

Barrientos, A. and Villa, J. M. (2015). Antipoverty transfers and labour market outcomes: Regression discontinuity design findings. *The Journal of Development Studies*, 51(9):1224–1240.

Blattman, C., Fiala, N., and Martinez, S. (2012). Employment generation in rural africa: mid-term results from an experimental evaluation of the youth opportunities program in northern uganda.

Blattman, C., Fiala, N., and Martinez, S. (2014). Generating skilled self-employment in developing countries: Experimental evidence from uganda. *The Quarterly Journal of Economics*, 129(2):697–752.

Blattman, C., Fiala, N., and Martinez, S. (2020). The long-term impacts of grants on poverty: Nine-year evidence from uganda's youth opportunities program. *American Economic Review: Insights*, 2(3):287–304.

Card, D., Kluve, J., and Weber, A. (2010). Active labour market policy evaluations: A meta-analysis. *The economic journal*, 120(548):F452–F477.

Card, D., Kluve, J., and Weber, A. (2018). What works? a meta analysis of recent active labor market program evaluations. *Journal of the European Economic Association*, 16(3):894–931.

Chaudhury, N. and Parajuli, D. (2010). Conditional cash transfers and female schooling: the impact of the female school stipend programme on public school enrolments in punjab, pakistan. *Applied Economics*, 42(28):3565–3583.

de Bem Lignani, J., Sichieri, R., Burlandy, L., and Salles-Costa, R. (2011). Changes in food consumption among the programa bolsa família participant families in brazil. *Public health nutrition*, 14(5):785–792.

Dona, G. J. (2023). Mothers' labor supply and conditional cash transfers: Evidence from chile.

Eissa, N. and Hoynes, H. W. (2006). Behavioral responses to taxes: Lessons from the eitc and labor supply. *Tax policy and the economy*, 20:73–110.

Fernald, L. C., Gertler, P. J., and Neufeld, L. M. (2008). Role of cash in conditional cash transfer programmes for child health, growth, and development: an analysis of mexico's oportunidades. *The lancet*, 371(9615):828–837.

Fernald, L. C., Gertler, P. J., and Neufeld, L. M. (2009). 10-year effect of oportunidades, mexico's conditional cash transfer programme, on child growth, cognition, language, and behaviour: a longitudinal follow-up study. *The Lancet*, 374(9706):1997–2005.

Fernandez, L. and Olfindo, R. (2011). Overview of the philippines' conditional cash transfer program: The pantawid pamilyang pilipino program (pantawid pamilya). *Philippine social protection note*, 2(2):1–12.

Fisher, J. and Rehkopf, D. H. (2022). The earned income tax credit as supplementary food benefits and savings for durable goods. *Contemporary Economic Policy*, 40(3):439–455.

Fruttero, A., Leichsenring, A. R., and Paiva, L. H. (2020). *Social programs and formal employment: Evidence from the Brazilian bolsa família program*. International Monetary Fund.

Gan, L. et al. (2020). The impact of labor income reward plan: An eitc experiment in china.

Garganta, S. and Gasparini, L. (2015). The impact of a social program on labor informality: The case of auh in argentina. *Journal of Development Economics*, 115:99–110.

Gerard, F., Naritomi, J., and Silva, J. (2021). Cash transfers and formal labor markets: Evidence from brazil.

Gertler, P. (2000). The impact of progresa on health.

Gertler, P., Boyce, S., et al. (2001). An experiment in incentive-based welfare: The impact of progresa on health in mexico. *University of California, Berkeley*, pages 30–37.

Goodman-Bacon, A. and McGranahan, L. (2008). How do eitc recipients spend their refunds? *Economic Perspectives*, 32(2).

Hanna, R. and Olken, B. A. (2018). Universal basic incomes versus targeted transfers: Anti-poverty programs in developing countries. *Journal of Economic Perspectives*, 32(4):201–226.

Hoddinott, J. and Skoufias, E. (2004). The impact of progresa on food consumption. *Economic development and cultural change*, 53(1):37–61.

Hotz, V. J. and Scholz, J. K. (2001). The earned income tax credit.

Hoynes, H. and Rothstein, J. (2016). Tax policy toward low-income families. Technical report, National Bureau of Economic Research.

Laguinge, L. A., Gasparini, L. C., and Neidhöfer, G. (2024). The long-run effects of conditional cash transfers: the case of bolsa familia in brazil. *Documentos de Trabajo del CEDLAS*.

List, J. A. (2020). Non est disputandum de generalizability? a glimpse into the external validity trial. Technical report, National Bureau of Economic Research.

Maluccio, J. and Flores, R. (2005). *Impact evaluation of a conditional cash transfer program: The Nicaraguan Red de Protección Social*. Intl Food Policy Res Inst.

Maluccio, J. A. (2005). Coping with the “coffee crisis” in central america: The role of the nicaraguan red de proteccion social.

Martins, A. P. B. and Monteiro, C. A. (2016). Impact of the bolsa família program on food availability of low-income brazilian families: a quasi experimental study. *BMC public health*, 16(1):827.

Masino, S. and Niño-Zarazúa, M. (2020). Improving financial inclusion through the delivery of cash transfer programmes: The case of mexico’s progresa-oportunidades-prospera programme. *The Journal of Development Studies*, 56(1):151–168.

McIntosh, C. and Zeitlin, A. (2024). Cash versus kind: Benchmarking a child nutrition program against unconditional cash transfers in rwanda. *The Economic Journal*, 134(664):3360–3389.

McIntosh, C. and Zeitlin, A. (2025). Skills and liquidity barriers to youth employment: Medium-term evidence from a cash benchmarking experiment in rwanda. *Journal of Human Resources*.

Meyer, B. D. and Rosenbaum, D. T. (2001). Welfare, the earned income tax credit, and the labor supply of single mothers. *The quarterly journal of economics*, 116(3):1063–1114.

Nichols, A. and Rothstein, J. (2016). The earned income tax credit” in economics of means-tested transfer programs in the united states.

Purba, R. A., Widianingsih, I., and Muhtar, E. A. (2023). The longlasting program keluarga harapan as conditional cash transfer in indonesia: The existing problems and the policy recommendations. *Journal La Sociale*, 4(5):426–433.

Qian, P. (2025). Market-oriented reforms and human capital reallocation in urban china: A gender perspective. *Labour Economics*, page 102811.

Ribas, R. P. and Soares, F. V. (2011). Is the effect of conditional transfers on labor supply negligible everywhere. *Unpublished manuscript, Tinker Fellowship, Center for Latin American and Caribbean Studies, University of Illinois at Urbana-Champaign.*

Samaniego, P. and Tejerina, L. (2010). Financial inclusion through the bono de desarrollo humano in ecuador. *Inter-American Development Bank, Technical Notes, Washington DC.*

Skoufias, E. (2001). Progresa and its impact on the human capital and welfare of household in rural mexico: A synthesis of the results of an evaluation by ifpri.

Skoufias, E. and McClafferty, B. (2001). Is progresa working? summary of the results of an evaluation by ifpri.

Todd, J. E., Winters, P. C., and Hertz, T. (2020). Conditional cash transfers and agricultural production: Lessons from the oportunidades experience in mexico. In *Migration, transfers and economic decision making among agricultural households*, pages 39–67. Routledge.

Ventry, D. J. (2000). The collision of tax and welfare politics: The political history of the earned income tax credit, 1969–99. *National Tax Journal*, pages 983–1026.

Vidigal, V. (2023). The impact of conditional cash transfers on labor supply: New evidence from brazil's bolsa familia. *Available at SSRN 4322137.*

Yuliani, E. and Nasrudin, R. a. (2024). The effects of the indonesian conditional cash transfer program on transition out of agriculture. *World Development*, 173:106425.

# Figures and Tables

| Stage | Date | Data |
|---|---|---|
| Pre-Pilot Survey | May, 2014 | Sample: 808 households (three-stage stratified random sampling)<br>Cross-Sectional Data |
| The Pilot | Dec, 2014 – May, 2015 | Sample: 59 households (27 randomly treated)<br>Panel Data: One Baseline + One Follow-up |
| The Experiment | Oct, 2015 – Jan, 2017 | Sample: 192 households (103 randomly treated)<br>Panel Data: One Baseline + Five Follow-ups |

Figure 1: Timeline (May, 2014 - Jan, 2017)

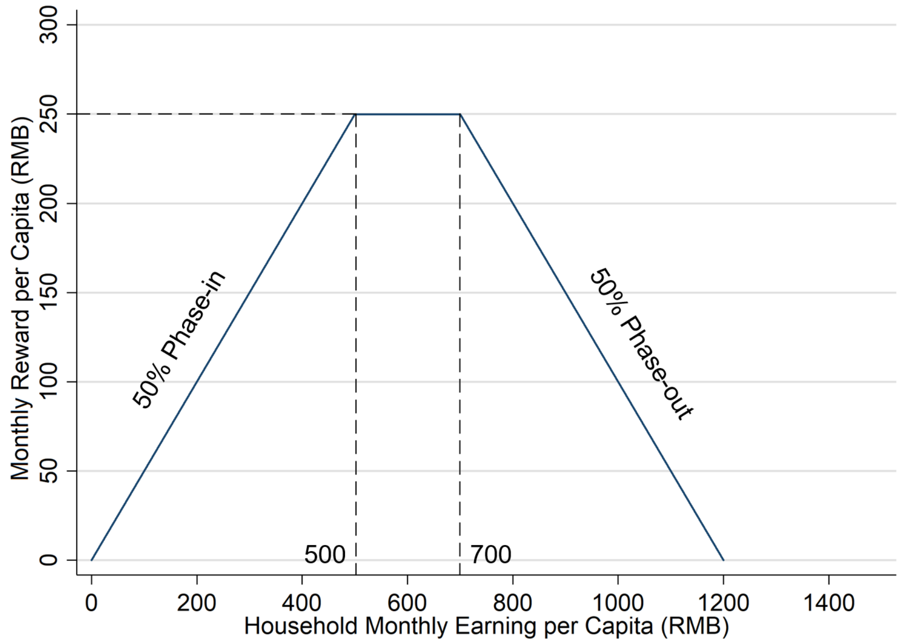



Figure 2: Labor Income Reward Plan (Pilot)

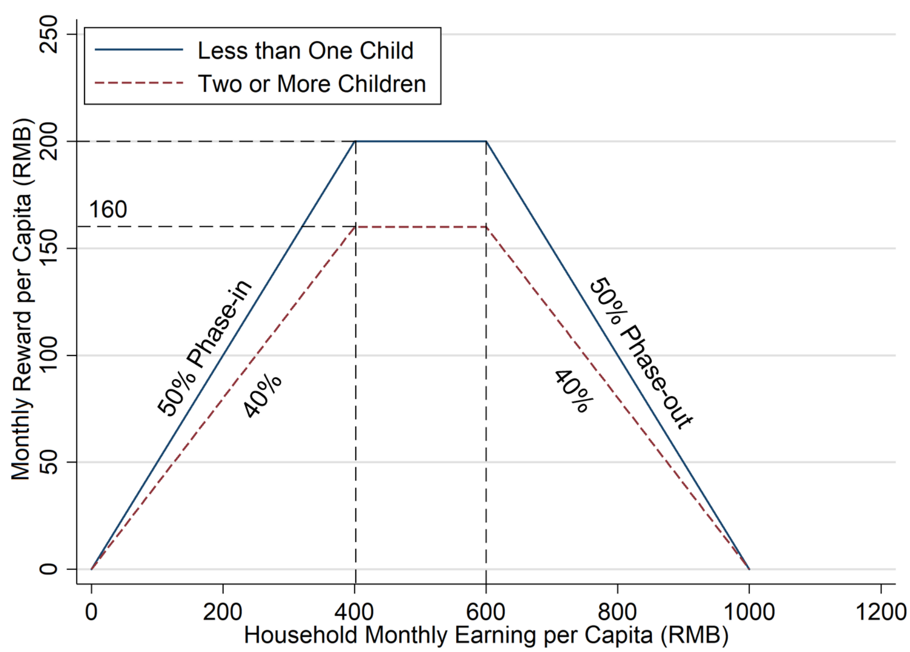



Figure 3: Labor Income Reward Plan (Experiment)


Figure 4: Reported Earning and Verified Earning

Note: Kernel density estimation, bandwidth=200.

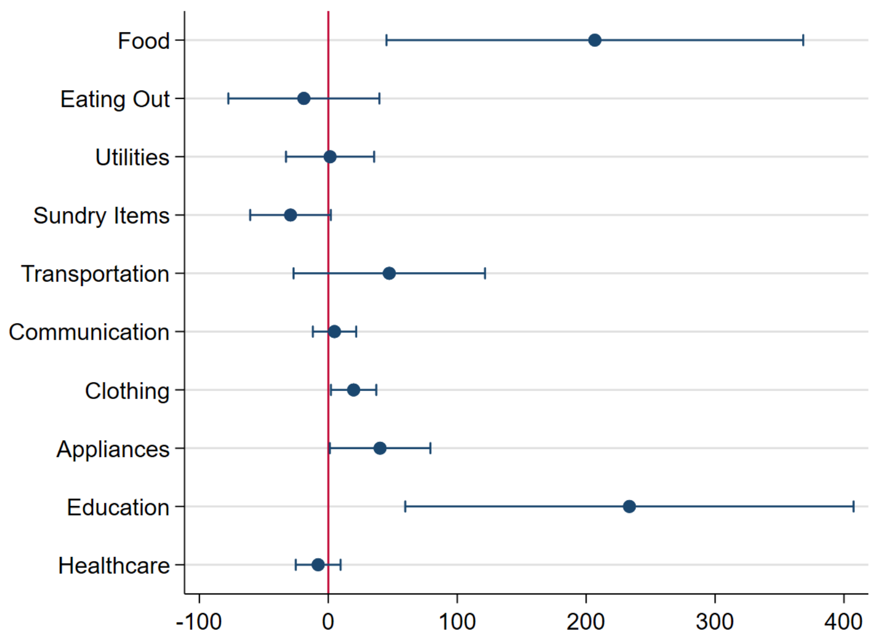



Figure 5: Effects on Household Monthly Consumption (Experiment)

Note: Estimated coefficients are presented along with 90% confidence intervals (capped spikes).

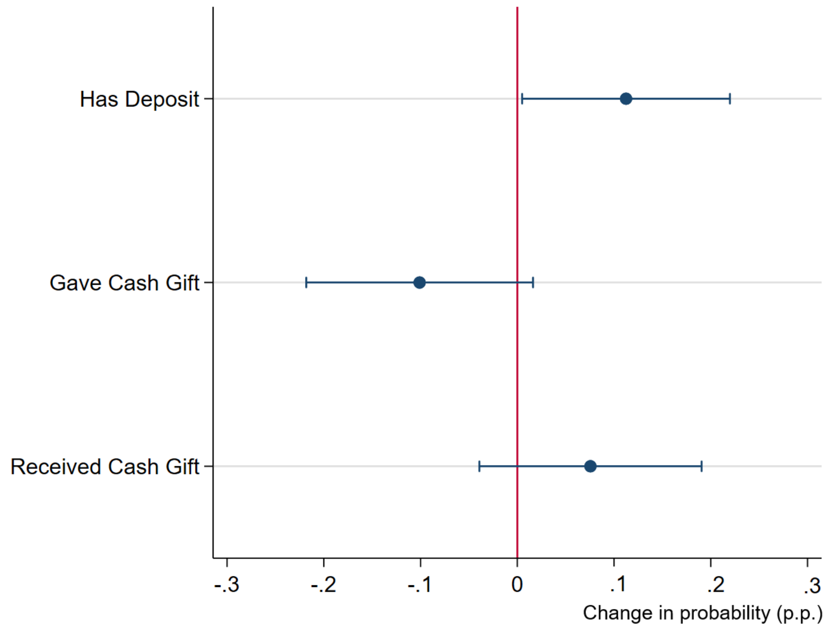



Figure 6: Effects on Saving and Inter-household Transfer

Note: Estimated coefficients are presented along with 90% confidence intervals (capped spikes).

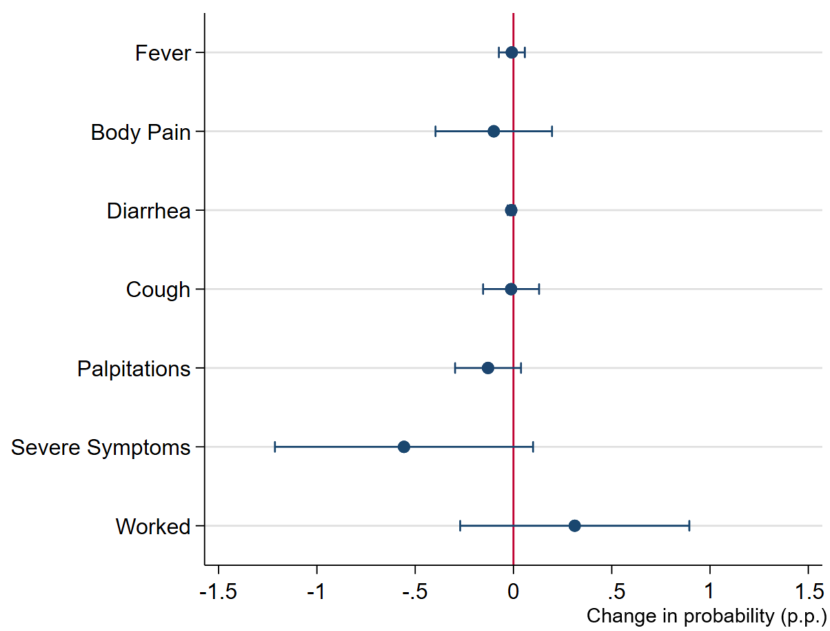



Figure 7: Effects on Health and Working through Food Consumption

Note: Estimated coefficients are presented along with 90% confidence intervals (capped spikes).

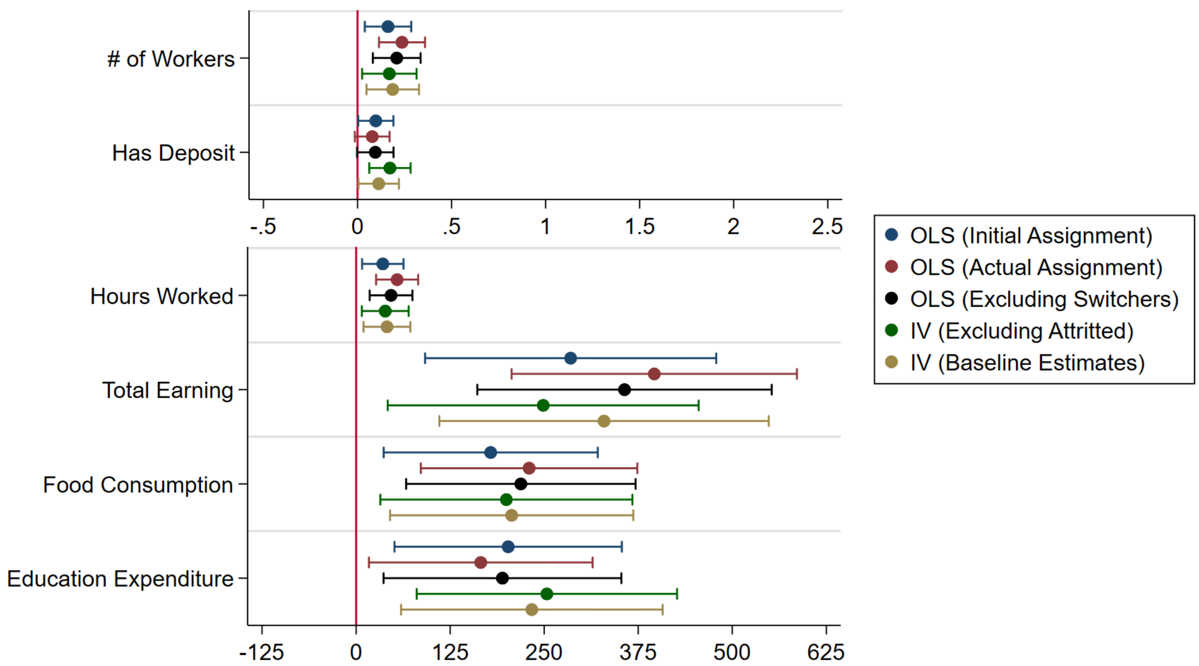



Figure 8: Robustness Checks

Note: Estimated coefficients are presented along with 90% confidence intervals (capped spikes).

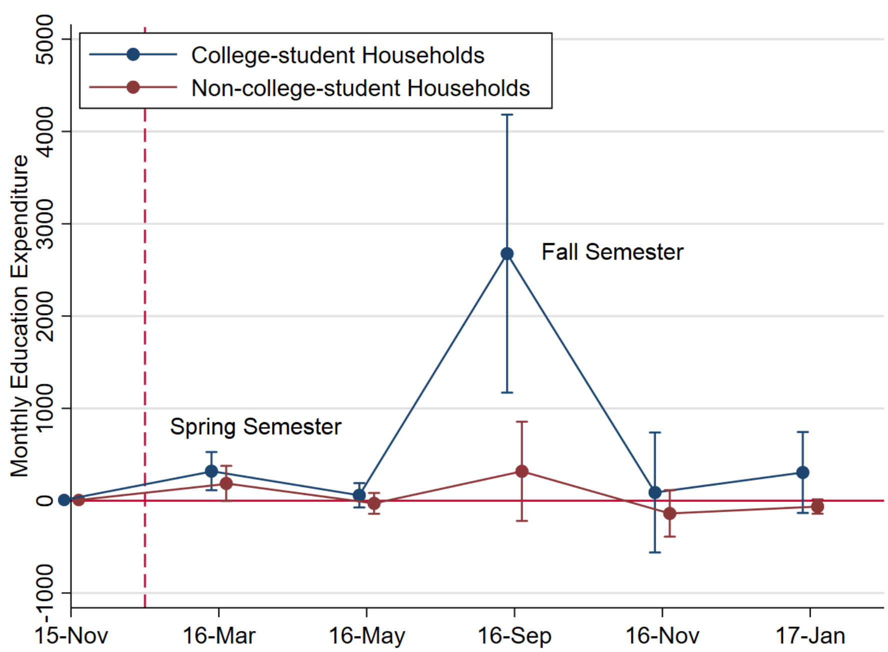



Figure 9: Dynamic Effects on Education Expenditure

Note: Estimated coefficients are presented along with 90% confidence intervals (capped spikes).

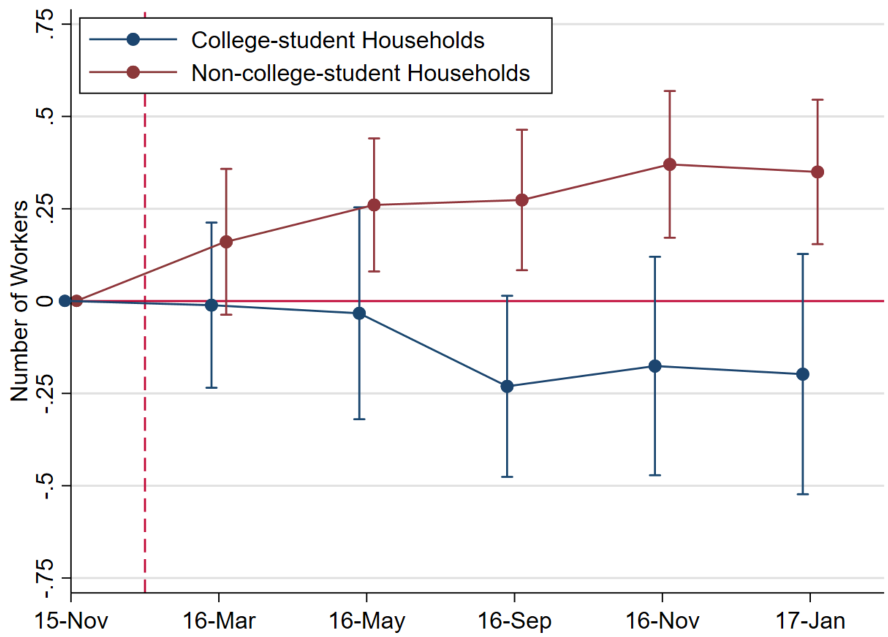



Figure 10: Effects on Number of Workers

Note: Estimated coefficients are presented along with 90% confidence intervals (capped spikes).

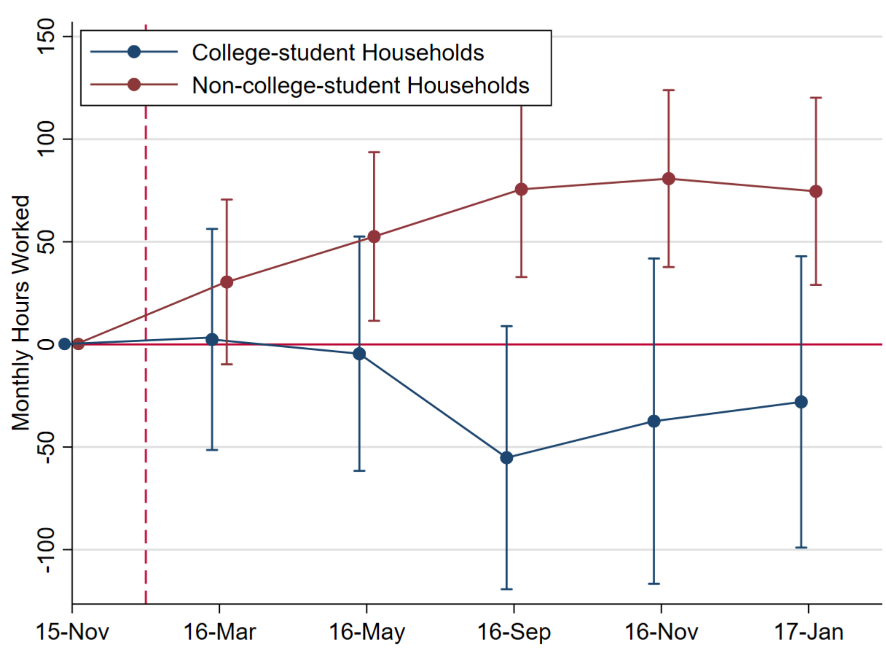



Figure 11: Effects on Monthly Hours Worked

Note: Estimated coefficients are presented along with 90% confidence intervals (capped spikes).

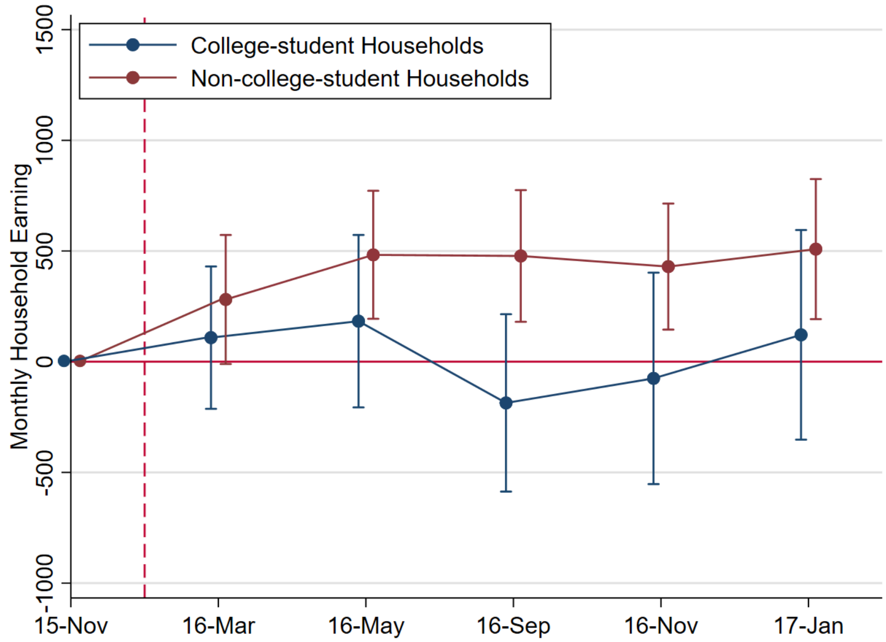



Figure 12: Effects on Monthly Household Earning

Note: Estimated coefficients are presented along with 90% confidence intervals (capped spikes).

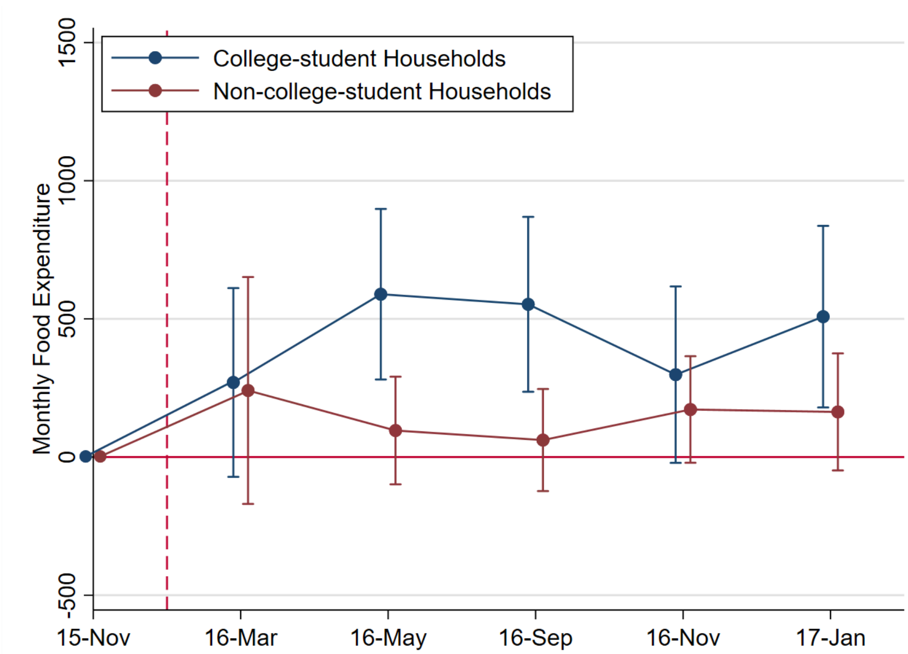



Figure 13: Heterogeneous Effects on Food Expenditure

Note: Estimated coefficients are presented along with 90% confidence intervals (capped spikes).

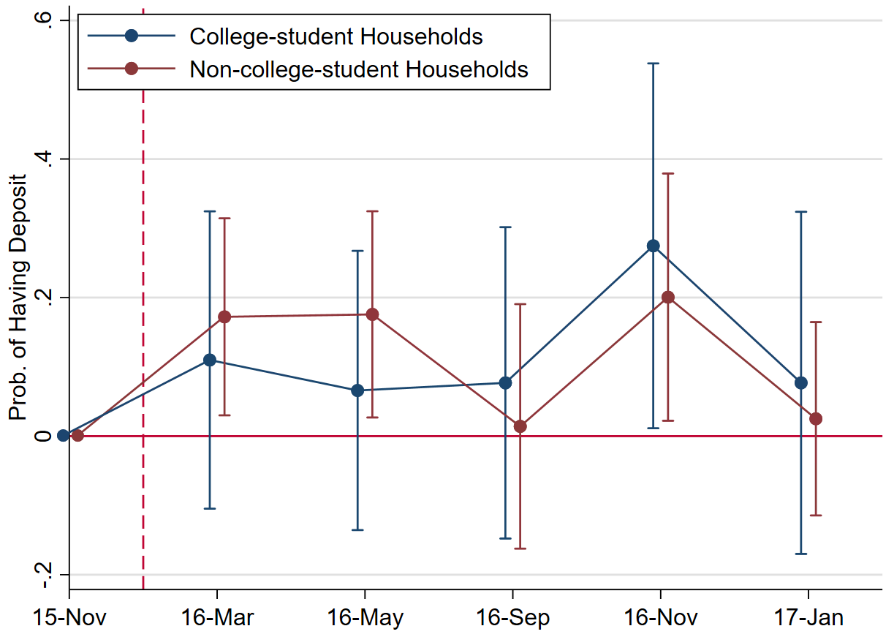



Figure 14: Heterogeneous Effects on Deposit

Note: Estimated coefficients are presented along with 90% confidence intervals (capped spikes).

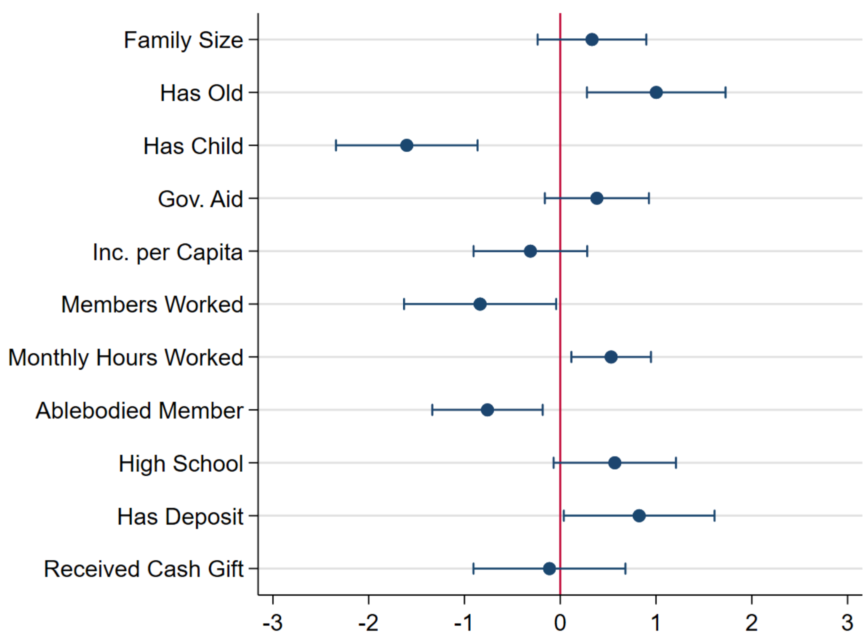



Figure 15: Factors Associated with College Household (Probit)

Note: Estimated coefficients are presented along with 90% confidence intervals (capped spikes).

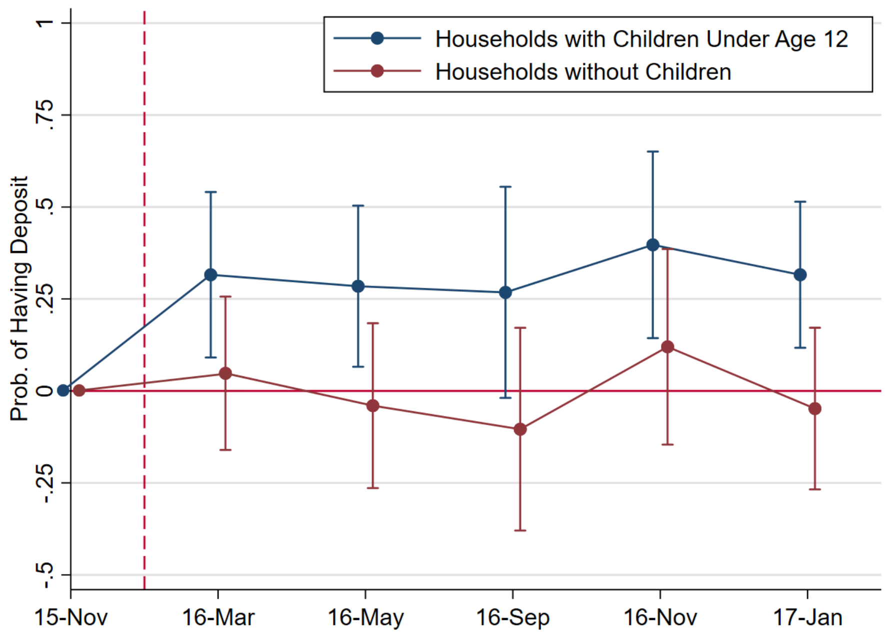
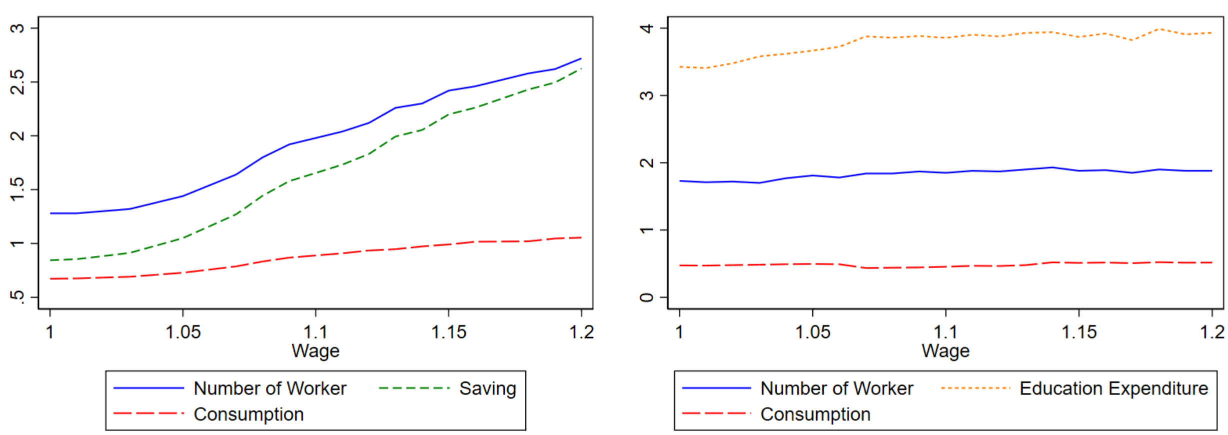



Figure 16: Heterogeneous Effects on Deposit

Note: Estimated coefficients are presented along with 90% confidence intervals (capped spikes).




(a) Non-college-student Households (b) College-student Households

Figure 17: Household's Average Response to Increasing Work Incentives

Parameters: $\rho = 0.3$, $\gamma = 1.9$, $\theta = 0.8$, $\alpha = 1.4$, $\theta = 0.9$, $\beta = 0.98$, $r = 1.05$.

Table 1: Impacts of CCT Programs on Labor Supply in Developing Context

| **Countries** | **Programs** | **Conditionalities** |
|---|---|---|
| Argentina | Universal Child Allowance (AUH) | Health checks + School attendance |
| Brazil | Programa de Erradicação do Trabalho Infantil (PETI) | No child labor + School attendance |
| | Bolsa Escola | School attendance |
| | Bolsa Alimentação | Immunisation + Health checks |
| | Bolsa Família | School attendance + Immunisation + Health checks |
| Bolivia | Bono Juancito Pinto (BJP) | School attendance |
| Chile | Unique Family Subsidy (SUF) | Health checks + School attendance |
| | Jefes y Jefas | Minimum hours of community work |
| Colombia | Familias en Acción | Health checks + School attendance |
| Ecuador | Bono de Desarrollo Humano | Health checks + School attendance |
| Honduras | Programa de Asignación Familiar (PRAF) | Health checks + School attendance |
| Indonesia | Program Keluarga Harapan (PKH) | Health checks + School attendance |
| | Timor Tenggah Utara Experiment | Migrate |
| Mexico | Progresa/ Oportunidades | Health checks + School attendance |
| Nicaragua | Red de Protección Social (RPS) | Health checks + School attendance |
| Pakistan | Punjab's Female School Stipend Program (FSSP) | School enrollment |
| Philippine | Pantawid Pamilyang Pilipino Program | Health checks + School attendance |
| Uganda | Youth Opportunities Program (YOP) | Formal advising |
| Uruguay | Plan de Atención Nacional a la Emergencia Social (PANES) | Health checks + School attendance |

Table 2: Balance of Covariate (The Pilot)

| | Treat | Control | T−C | p-value |
|---|---|---|---|---|
| ***Ablebodied*** | | | | |
| **Female** | 0.453 | 0.554 | -0.101 | 0.240 |
| | (0.502) | (0.500) | (0.086) | |
| **Age** | 46.5 | 48.2 | -1.6 | 0.530 |
| | (14.1) | (15.8) | (2.5) | |
| **Education**[1] | 5.85 | 4.51 | 1.34 | 0.100 |
| | (5.01) | (4.53) | (0.81) | |
| **Worked** | 0.531 | 0.595 | -0.063 | 0.458 |
| | (0.503) | (0.494) | (0.085) | |
| **Monthly Hours Worked** | 102.8 | 119.8 | -16.9 | 0.412 |
| | (115.5) | (125.6) | (20.6) | |
| **Monthly Earning** | 589.3 | 613.8 | -24.4 | 0.852 |
| | (814.8) | (729.1) | (131.4) | |
| ***Household*** | | | | |
| **Family Size** | 3.51 | 3.09 | 0.42 | 0.205 |
| | (1.34) | (1.20) | (0.33) | |
| **Has Old/Child** | 0.630 | 0.625 | 0.005 | 0.971 |
| | (0.492) | (0.492) | (0.129) | |
| **Recieve Gov. Aid**[2] | 0.037 | 0.031 | 0.006 | 0.905 |
| | (0.192) | (0.177) | (0.048) | |
| **Has Urban Register**[3] | 1.000[4] | 1.000 | 0.000 | 1.000 |
| | (0.000) | (0.000) | (0.000) | |
| **Monthly Inc. per Capita** | 710.1 | 873.9 | -163.7 | 0.391 |
| | (415.9) | (906.7) | (189.5) | |
| **Members Worked** | 1.25 | 1.37 | -0.11 | 0.561 |
| | (0.65) | (0.83) | (0.19) | |
| **Monthly Hours Worked** | 243.7 | 277.0 | -33.3 | 0.476 |
| | (163.1) | (189.5) | (46.5) | |
| **Monthly Earning** | 1,397.0 | 1,419.5 | -22.5 | 0.942 |
| | (1,196.1) | (1,149.8) | (306.0) | |
| **# of Ablebodied** | 74 | 64 | **Total** | 183 |
| **# of Individuals** | 95 | 99 | **Total** | 194 |
| **# of Households** | 27 | 32 | **Total** | 59 |

[1] Education = Years of education
[2] Gov. Aid = Household receive any government aid (e.g. Dibao)
[3] Urban = Household members hold urban household register (Hukou)
[4] There is no rural household in the pilot study
Table presents the balance of covariates between the treatment and the control groups in the pilot. The last column shows the p-values of two-sample t-tests. We find no statistical difference in either individual or household characteristics between the two groups. The randomization is successful.

## Table 3: Balance of Covariates (The Experiment)

| | Initial Assignment | | | | Actual Assignment | | | |
|---|---|---|---|---|---|---|---|---|
| | **Treat** | **Control** | **T−C** | **p-value** | **Treat** | **Control** | **T−C** | **p-value** |
| ***Ablebodied*** | | | | | | | | |
| **Female** | 0.563 | 0.520 | 0.043 | 0.421 | 0.549 | 0.534 | 0.015 | 0.778 |
| | (0.498) | (0.501) | (0.054) | | (0.499) | (0.500) | (0.054) | |
| **Age** | 44.8 | 45.7 | -0.9 | 0.543 | 44.4 | 46.3 | -1.8 | 0.218 |
| | (14.5) | (13.9) | (1.5) | | (14.2) | (14.1) | (1.5) | |
| **Education**[1] | 7.11 | 6.63 | 0.48 | 0.237 | 7.03 | 6.68 | 0.35 | 0.382 |
| | (3.90) | (3.67) | (0.40) | | (3.84) | (3.72) | (0.40) | |
| **Worked** | 0.473 | 0.480 | -0.007 | 0.891 | 0.462 | 0.497 | -0.035 | 0.516 |
| | (0.501) | (0.501) | (0.054) | | (0.500) | (0.502) | (0.054) | |
| **Monthly Hrs. Worked** | 71.6 | 81.9 | -10.3 | 0.350 | 69.3 | 87.1 | -17.8 | 0.108 |
| | (99.4) | (105.0) | (11.0) | | (97.8) | (108.3) | (11.0) | |
| **Monthly Earning** | 437.7 | 521.9 | -84.1 | 0.260 | 429.1 | 547.4 | -118.2 | 0.114 |
| | (663.4) | (719.8) | (74.5) | | (654.5) | (735.6) | (74.6) | |
| ***Household*** | | | | | | | | |
| **Family Size** | 2.78 | 2.78 | 0.00 | 0.981 | 2.75 | 2.79 | -0.04 | 0.800 |
| | (1.19) | (1.00) | (0.15) | | (1.19) | (0.99) | (0.16) | |
| **Has Old/Child** | 0.511 | 0.546 | -0.036 | 0.623 | 0.505 | 0.551 | -0.046 | 0.530 |
| | (0.503) | (0.500) | (0.073) | | (0.502) | (0.500) | (0.073) | |
| **Gov. Aid**[2] | 0.691 | 0.649 | 0.042 | 0.549 | 0.709 | 0.629 | 0.080 | 0.255 |
| | (0.464) | (0.501) | (0.070) | | (0.457) | (0.509) | (0.070) | |
| **Urban**[3] | 0.968 | 0.938 | 0.030 | 0.332 | 0.961 | 0.944 | 0.017 | 0.573 |
| | (0.177) | (0.242) | (0.031) | | (0.194) | (0.232) | (0.031) | |
| **Monthly Inc. per Capita** | 489.4 | 571.0 | -81.5 | 0.139 | 475.9 | 609.8 | -133.9** | 0.018 |
| | (351.7) | (404.3) | (54.9) | | (351.2) | (425.9) | (56.0) | |
| **Members Worked** | 0.84 | 0.88 | -0.04 | 0.662 | 0.82 | 0.91 | -0.08 | 0.420 |
| | (0.69) | (0.76) | (0.10) | | (0.69) | (0.76) | (0.10) | |
| **Monthly Hrs. Worked** | 127.2 | 151.1 | -23.9 | 0.248 | 123.8 | 159.6 | -35.7* | 0.084 |
| | (131.7) | (152.3) | (20.6) | | (130.9) | (154.4) | (20.5) | |
| **Monthly Earning** | 777.7 | 963.2 | -185.4 | 0.175 | 766.7 | 1,002.5 | -235.8* | 0.083 |
| | (885.2) | (990.4) | (136.0) | | (884.0) | (993.9) | (135.5) | |
| **# of Ablebodied** | 179 | 168 | **Total** | 347 | 163 | 184 | **Total** | 347 |
| **# of Individuals** | 262 | 270 | **Total** | 533 | 284 | 249 | **Total** | 533 |
| **# of Households** | 95 | 97 | **Total** | 192 | 103 | 89 | **Total** | 192 |

[1] Education = Years of education
[2] Gov. Aid = Household receive any government aid (e.g. Dibao)
[3] Urban = Household members hold urban household register (Hukou)
Table presents the balance of covariates between the treatment and the control groups in the experiment. The last column in each panel shows the p-values of two-sample t-tests. The right panel compares the treatment and the control groups by actual assignment. Because some of the assignments were switched at the request of the local government, the treatment group is relatively poor (see the income per capita in the last row. However, the left panel shows that the initial assignment is reasonably balanced and therefore could be used as the instrumental variable.

Table 4: Treatment Effects on Household Labor Supply and Earning

| | # of Workers | Monthly Total Hours Worked | Monthly Total Earning |
|---|---|---|---|
| **Panel A: Household Labor Supply and Earning (The Pilot)** | | | |
| **treated** | 0.35* | 95.64** | 349.73 |
| | (0.20) | (42.81) | (245.93) |
| **Household FE** | Yes | Yes | Yes |
| **Time FE** | Yes | Yes | Yes |
| **# of Household** | 63 | 63 | 63 |
| **N** | 112 | 112 | 112 |
| **Panel B: Household Labor Supply and Earning (The Experiment)** | | | |
| | **# of Workers** | **Monthly Total Hours Worked** | **Monthly Total Earning** |
| **treated** | 0.19** | 41.03** | 329.53** |
| | (0.08) | (18.92) | (133.06) |
| **First-stage F** | | 589.49 | |
| **Prob. > F** | | 0.000 | |
| **Household FE** | Yes | Yes | Yes |
| **Time FE** | Yes | Yes | Yes |
| **# of Household** | 191 | 191 | 191 |
| **N** | 1,146 | 1,146 | 1,146 |

*** p<0.01, ** p<0.05, * p<0.1; Standard errors (in parentheses) are clustered on household-level.

Table shows the estimates of treatment effects on household-level outcomes. Panel A shows that the pilot treatment significantly increased households' numbers of worked members, total hours worked in the previous month. Panel B shows that the experiment treatment significantly increased numbers of workers, total hours worked, and total earnings in the previous month. Finally, the large Kleibergen–Paap rk Wald F statistic indicates strong instrument relevance in the first stage.

Table 5: Results Summary

| | Pilot | Experiment | College Household | Non-college Household |
|---|---|---|---|---|
| **Worked more?** | Yes | Yes | No | Yes |
| **Earned more?** | No | Yes | No | Yes |
| **Spent more?** | - | Yes | Yes | No |
| **Saved more?** | - | Yes | No | Yes |
| **Food Improved health?** | - | No | - | - |
| **Food improved productivity?** | - | No | - | - |
| **Borrow/Lend more?** | - | No | - | - |

Table 6: Balance of Age Groups

| | Treat | Control | T-C | p-value |
|---|---|---|---|---|
| **7≤age≤12** | 0.064 | 0.021 | 0.043 | 0.138 |
| **(Primary School)** | (0.246) | (0.143) | (0.029) | |
| **12≤age≤15** | 0.043 | 0.072 | -0.030 | 0.382 |
| **(Junior High)** | (0.203) | (0.260) | (0.034) | |
| **15≤age≤18** | 0.096 | 0.082 | 0.013 | 0.749 |
| **(Senior High)** | (0.296) | (0.277) | (0.041) | |
| **18≤age≤22** | 0.202 | 0.144 | 0.058 | 0.293 |
| **(College)** | (0.404) | (0.353) | (0.055) | |
| **N** | 94 | 97 | | |

Table presents the age distribution and corresponding education stages (in parentheses) for children in the treatment and control groups. Specifically, we construct indicator variables for the presence of children within each age group. The first two columns of the table report the means and standard deviations (in parentheses) of these variables, the third column reports the mean differences and pooled standard deviations from a two-sample t-test, and the last column reports the associated p-values. The results show no meaningful differences between the treatment and control groups across education stages, suggesting that the observed effect on education expenditure is not driven by differences in child composition between the two groups.

Table 7: Heterogeneous Treatment Effects on Labor Supply and Saving

| | **# of Workers** | **# of Workers** | **Has Deposit** |
|---|---|---|---|
| **treated** | 0.147 | -0.102 | 0.298*** |
| | (0.090) | (0.082) | (0.052) |
| **treated×phase-in** | 0.095 | | |
| | (0.113) | | |
| **treated×available-worker**[1] | | 0.477*** | |
| | | (0.090) | |
| **treated×already-saved** | | | -0.837*** |
| | | | (0.088) |
| **Household FE** | Yes | Yes | Yes |
| **Time FE** | Yes | Yes | Yes |
| **# of Household** | 191 | 191 | 191 |
| **N** | 1,146 | 1,146 | 1,146 |

*** $p<0.01$, ** $p<0.05$, * $p<0.1$; Standard errors (in parentheses) are clustered on household-level.

[1] Available-worker = households with idle workers (able-bodied members who are not working).

Table reports IV estimates of the treatment effects on household labor supply, earnings, and savings. The first column shows that households in the phase-in region do not differ in their labor-supply response, indicating that heterogeneity between college-student and non-college-student households cannot be attributed to differences in incentives alone. The second column shows that households with non-working able-bodied members at baseline significantly increased the number of workers, consistent with college-student households' limited labor capacity. The third column shows that the saving effect is substantially attenuated among households already saving at baseline, explaining the absence of a saving response among college-student households.